1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L5 – Usability Requirement**

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3 **Recap**

- What we learned so far
  - General introduction to the field
  - Software engineering life cycles for interactive systems

4 **Recap - ISDLC**

5 **Note**

- First stage: feasibility study (we'll not spend much time on it)
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- Instead, we'll discuss the second stage – requirement gathering, analysis and specifications
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7 **Recap**

- System Centered Design (SCD) vs User Centered Design (UCD)
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8 **System Centered Design**

9 **System Centered Design**

- Main concerns
  - What can I easily build on this platform?
  - What can I create from the available tools?
  - What do I as a programmer find interesting?

10 **User Centered Design**

- Design is based upon a user's
  - Abilities
  - Context

- Goals/needs
- Not only programmer's convenience

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12 **Example Scenario**

- Recall the calendar app (mobile app)
- What we mean by requirement gathering, analysis & specification for this system
  — What are the requirements?

13 **Example Scenario**

- "We need to develop it for mobile platforms" – is it part of "requirement"?
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- "What programming language are we going to use – Java, Java scripts, …" – is it a "requirement"?
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- "How we are going to store the various information needed (e.g., reminders, special events); should we use some RDBMS tools (e.g., my SQL)" – is it a "requirement"?
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14 **Example Scenario**

- There may be many more such questions, which may come or our mind regarding "requirements"
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15 **Example Scenario**

- These are NOT requirements per se

16 **Example Scenario**

- Some are related to "feasibility"
  - "We need to develop it for mobile platforms"
  - "What programming language are we going to use – Java, Java scripts, …"

17 **Example Scenario**

- Some may be related to "design"
  - "How we are going to store the various information needed (e.g., reminders, special events); should we use some RDBMS tools (e.g., my SQL)"

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19 **Our Objective**

- To identify requirements that
  - Ensures usability
  - Leads to system specification (for design and implementation of the system)

20 **What We Need**

- Gathering of usability requirements
- Analysis and specification of usability requirements (to be able to convert those to system design)

21 **What We Need**

- Gathering of usability requirements – generally treated as a type of "non-functional requirement"

22 **Non-Functional Requirements**

23 **NFR - What**

- These are system characteristics that can not be expressed as functions
  - Related to quality attributes of the software

24 **Broad Categories**

- Performance related
- Operating constraints
- Economic considerations
- Life-cycle requirements
- Interface issues
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26 **NFR – Performance Related**

- Reliability – how to ensure
- Security – what are the things required
- Response time – ideal …

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28 **NFR – Operating Constraints**

- Physical constraints (size)
- Personnel availability
- Skill level considerations
- System accessibility for maintenance …

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30 **NFR – Economic Considerations**

- Immediate and/or long-term costs for design, development and implementation

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32 **NFR – Life-Cycle Requirements**

- Design quality - measured in terms of maintainability, enhance ability, portability …

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34 **NFR – Interface Issues**

- How system is to interface with its environment, users, and other systems
  - External system interface (H/W & S/W)
  - Usability (out focus in this course)

35 **NFR - Usability**

- Remember we are dealing with interactive systems – meant to be used by "non-

experts”
- Usability very important

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37 **Specifying NFR - Advantages**
- Ensure software follows legal and adherence rules
- Ensure reliability, availability, performance, and scalability of the system
- Help constructing security policy
- Ensure good user experience & ease of operating the software

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39 **Not Specifying NFR – Consequences!**
- May lead to unsatisfied users, clients, and developers
- May result in inconsistent software
- There may be time and cost overrun to fix software later, to incorporate NFR

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41 **Note**
- In the next lecture, we shall learn how to gather usability requirements
- And also how to specify!
  - Generally termed as “functional” requirement specification
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42 **Reference**

- Rajib Mall (2018). Fundamentals of Software Engineering, 5th ed, PHI Learning Pvt Ltd. Chapter 4
- Roger S Pressman (2020). Software Engineering: A Practitioner's Approach, 9th ed, McGraw-Hill Education, New York, Chapter 7-8

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